

PROTOTYPING AND DESIGN VALIDATION STUDIO

# AI Presenter or Avatar Video

*A 45-minute, click-by-click CampusConnect prototype using HeyGen, ChatGPT, Gradio, and GitHub*

|            |             |                      |            |
|------------|-------------|----------------------|------------|
| CLASS TIME | NEW AI TOOL | GENERATION           | VALIDATION |
| 45 minutes | HeyGen      | 1 stock-avatar video | 5 tests    |

REQUIRED PATH

Every student creates the same 30-60 second HeyGen Quick Avatar Video with a public stock avatar. Do not create a personal digital twin in this graded lab. The video becomes an optional orientation layer beside the existing local Gradio chatbot; it does not replace RAG, text, or human help.

## The fictional change request

The Week 7 voice prototype helped CampusConnect users ask a question without typing, but a Monday-morning usability review exposed a new uncertainty. Several first-time users did not understand whether the voice was a live staff member, what topics CampusConnect could answer, or what would happen when the approved guide did not contain an answer. One user also kept the computer muted in a public study area and missed the spoken opening entirely.

The team proposes a short AI presenter that explains the chatbot before the first question. The presenter will name the supported IT topics, warn users not to share private information, and explain the human fallback. The design team does not yet know whether the video improves comprehension or merely adds clutter, delay, and another dependency. Therefore, Week 8 builds a low-cost prototype and gathers evidence before Week 9 implementation and code review.

The architect's decision is intentionally reversible. The video must be optional, must not autoplay, and must have a complete written transcript. If the presenter is blocked, muted, slow, or ignored, the existing text chatbot must still work. The team will keep, change, or remove the feature based on five observable tests.

## What you will produce

| Deliverable         | Exact result                                                                                                                          |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| HeyGen prototype    | One 30-60 second landscape video named CampusConnect Welcome - <Your Last Name>, using a public stock avatar and the supplied script. |
| Gradio interface    | An AI Presenter card in the existing local app with disclosure, video embed or safe share-link fallback, transcript, and no autoplay. |
| Validation evidence | Five completed tests plus one KEEP / CHANGE / REMOVE design decision.                                                                 |
| GitHub trail        | Updated app.py and docs/prototypes/week-8/VIDEO_VALIDATION.md on feature/week-8-avatar-prototype, submitted through one draft PR.     |

### SAME GRADED WORK FOR EVERY STUDENT

Every student uses a stock avatar, the same script, the same five tests, and the same GitHub paths. Students may help one another find a menu, but there is no team-lead-only action and no separate personal-digital-twin track.

## The 45-minute clock

| Part              | Minutes | Stop when...                                     |
|-------------------|---------|--------------------------------------------------|
| 1. Open the trail | 5       | The Week 8 branch and validation file exist      |
| 2. Generate       | 12      | One HeyGen stock-avatar video is processing      |
| 3. Integrate      | 13      | The presenter card and text chatbot open locally |
| 4. Validate       | 9       | All five tests and one decision are recorded     |
| 5. Preserve       | 6       | Files are committed and one draft PR is open     |

Free-plan guardrail: [HeyGen currently advertises up to three free videos per month, each up to one minute](#). Availability can vary by account or region. This lab spends exactly one generation and reuses that same project for the final demonstration. Do not upgrade, create test generations, or delete the finished project.

## Before the 45-minute clock

- ☐ Your Week 4 campusconnect-gradio-spike folder opens and app.py is present.
- ☐ LM Studio and the local RAG app worked previously, or you have the saved Week 4 app.py.
- ☐ You can sign in to your individual GitHub semester repository.
- ☐ You have a free HeyGen account. Creating the account early does not consume a video.

- ☐ No real campus record, student data, password, recovery code, or API key will be used.

## Part 1 - Open the GitHub trail (5 minutes)

### TOOLS

Use the GitHub website. The branch and validation file make the prototype traceable before the new media or code is created.

### Step 1 - Create the Week 8 branch

1. Open <https://github.com/>, sign in, and open the same individual CampusConnect repository used in Weeks 1, 3, 4, and 7.
2. On the repository Code page, click the branch selector that usually shows main. Type feature/week-8-avatar-prototype.
3. Click Create branch: feature/week-8-avatar-prototype from main. Confirm that the branch selector now shows the Week 8 branch.
4. If the menu looks different, open [GitHub's official branch instructions](#).

### Step 2 - Create the validation record

1. While feature/week-8-avatar-prototype is selected, click Add file, then Create new file.
2. Enter docs/prototypes/week-8/VIDEO\_VALIDATION.md in the filename box. GitHub creates the folders from the forward slashes.
3. Paste Block A and then Block B into the editor. Replace <YOUR NAME>, <DATE>, and <YOUR LAST NAME>. Leave the test observations blank.
4. Click Commit changes. Use the message Start Week 8 video validation and commit to the current Week 8 branch.

**COPY / PASTE - VIDEO\_VALIDATION.md - Block A**

# Week 8 AI Presenter Prototype and Validation

Student: <YOUR NAME>

Date: <DATE>

Branch: feature/week-8-avatar-prototype

HeyGen project: CampusConnect Welcome - <YOUR LAST NAME>

### ## 1. Prototype hypothesis

If a short AI presenter explains the chatbot's scope, privacy boundary, and human fallback before the first question, a new user will understand how to begin without making the text path harder to use.

### ## 2. Required prototype behavior

- The presenter is a HeyGen public stock avatar, not an impersonation or clone.
- The visible disclosure says the presenter is AI-generated.
- The video does not autoplay.
- A complete transcript is visible without playing the video.
- The existing text chatbot remains usable if the video is paused, muted, blocked, slow, or ignored.
- The script contains no real student, employee, or institutional private data.
- The original local RAG grounding, citations, abstention, and human fallback remain unchanged.

**COPY / PASTE - VIDEO\_VALIDATION.md - Block B (paste after Block A)**

### ## 3. Five-test record

| # | Test                                              | Expected behavior                                                            | Observed result | Pass / Revise |
|---|---------------------------------------------------|------------------------------------------------------------------------------|-----------------|---------------|
| 1 | Open the page but do not play the video.          | AI disclosure and transcript are visible.                                    | <FILL IN>       | <FILL IN>     |
| 2 | Play the presenter once.                          | It is 30-60 seconds and states scope, privacy, and fallback.                 | <FILL IN>       | <FILL IN>     |
| 3 | Mute or pause the video.                          | The transcript and text chatbot still communicate the essential information. | <FILL IN>       | <FILL IN>     |
| 4 | Ask: How do I reset my CampusConnect password?    | The existing RAG answer and source citation still work.                      | <FILL IN>       | <FILL IN>     |
| 5 | Ask: What are the Thanksgiving dining hall hours? | The chatbot abstains or gives a safe human next step.                        | <FILL IN>       | <FILL IN>     |

### ## 4. One design decision

- Evidence I observed:
- Decision: KEEP / CHANGE / REMOVE the presenter path
- One reason:
- One small revision made:

### ## 5. Engineering and AI Use Log

- AI tools used: HeyGen and ChatGPT
- What HeyGen generated:
- What ChatGPT changed in app.py:
- What I verified myself:
- Remaining limitation:

## What is being tested?

| Weak claim                                | Testable prototype hypothesis                                                                                  |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| An avatar will make CampusConnect better. | A 30-60 second optional presenter will clarify scope, privacy, and fallback without blocking the text chatbot. |
| Users like video.                         | A muted user can understand the same opening through the disclosure and transcript without pressing Play.      |

## Part 2 - Generate one stock-avatar video (12 minutes)

### TOOL

Use HeyGen Quick Avatar Video. It is faster than the full AI Studio and keeps the class focused on the prototype hypothesis rather than video editing.

### Step 3 - Sign in and open Quick Avatar Video

1. Open the HeyGen web app at <https://app.heygen.com/>. If you need a free account, use <https://auth.heygen.com/signup>. Do not enter payment information or start a trial.
2. In the left navigation, open Apps, choose All Apps, and select Quick Avatar Video.
3. Keep HeyGen's official [Quick Avatar Video guide](#) open in another tab. It shows the current sequence: choose an avatar, add a script, choose a voice and motion engine, then generate.

### Step 4 - Choose the deliberately simple settings

1. Under Public Avatars, select one stock presenter that looks clear and appropriate for a university help experience. Do not upload a photo or create a personal digital twin.
2. Choose Landscape (16:9). This orientation fits beside or above a desktop chatbot more naturally than a vertical phone video.
3. Choose one clear preset voice. Preview no more than two voices. Leave language, speed, accent, expression, and other advanced controls at their defaults.
4. Choose Avatar III as the motion engine. It is the fastest option and is sufficient for a low-fidelity design-validation prototype.
5. Choose 720p if HeyGen asks for a resolution. Do not open Edit in Studio; the full editor adds choices that are not needed today.

| Use in this lab     | Do not use in this lab                            | Why                                                                 |
|---------------------|---------------------------------------------------|---------------------------------------------------------------------|
| Public stock avatar | Personal digital twin or uploaded classmate photo | The graded path stays identical and avoids consent/training delays. |

| Use in this lab      | Do not use in this lab                                     | Why                                                                                   |
|----------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Avatar III defaults  | Advanced gestures, background generation, or scene editing | The uncertainty is whether the orientation helps, not whether the video is cinematic. |
| One final generation | Several voice or appearance experiments                    | The free quota must remain available for the semester demo.                           |

## Step 5 - Paste the exact script and generate

1. Click Add Script. Delete any sample text, then paste the exact CampusConnect script below.
2. Name the project CampusConnect Welcome - <Your Last Name>. Read the script once before generating. It should disclose AI, name the scope, protect privacy, explain fallback, and mention the transcript.
3. Click Generate Video. Confirm only one generation. While HeyGen processes the video, move immediately to Part 3; do not wait on this screen.

### **COPY / PASTE - HeyGen presenter script**

Hello. I am an AI-generated CampusConnect presenter, not a live staff member. You can use the chatbot beside me for approved campus IT help.

Ask about password resets, secure campus Wi-Fi, or the IT Service Desk. Please do not share passwords, recovery codes, student records, or other private information.

If the chatbot cannot support an answer from its approved guide, it will direct you to a human. You may keep this video paused or muted and read the transcript below.

### **WHY THE PRESENTER IS PRERECORDED**

This is not a live avatar and it does not answer questions. The prerecorded presenter orients the user; the local RAG chatbot still performs retrieval and answer generation. Keeping those responsibilities separate makes the prototype cheap, testable, and replaceable.

## Part 3 - Add the presenter path to CampusConnect (13 minutes)

### TOOLS

Use HeyGen for the share or embed source, ChatGPT for one constrained code edit, and VS Code to run the existing local Gradio application.

### Step 6 - Recover the video source without spending another generation

1. Return to HeyGen. Open Projects if necessary, then open CampusConnect Welcome - <Your Last Name>. Wait until the status says Completed or the video can play.
2. Click Share. Enable sharing if HeyGen asks. If an Embed option is available, copy the complete iframe embed code. Otherwise, click Copy link and copy the normal share URL.
3. Paste the copied text into a temporary plain-text note. Do not copy an API key, browser token, private dashboard URL, or edit link.
4. HeyGen explains its share and embed choices in [its official sharing guide](#). A normal share link is an acceptable fallback if the free account does not expose embeddable iframe code.

#### PROTOTYPE FALLBACK IS INTENTIONAL

An embedded player is preferable, but a clearly labeled share link beside the chatbot is still valid evidence. The code prompt must not fabricate an embed address from a normal link. Record which path your free account allowed.

### Step 7 - Open and protect the existing local app

1. Open [Visual Studio Code](#), then choose File > Open Folder and open the existing campusconnect-gradio-spike folder from Week 4.
2. In Explorer, confirm app.py and requirements.txt are present. Open app.py and choose File > Save As. Save one safety copy named app\_week4\_backup.py in the same folder.
3. Open a VS Code terminal and run python app.py once. Open the local URL, usually <http://127.0.0.1:7860>, and confirm the text chatbot still opens. Stop the app with Control+C before editing.
4. If Gradio is unfamiliar, keep the official [Gradio quickstart](#) open for the local run pattern.

## Step 8 - Ask ChatGPT for one complete, constrained edit

1. Open <https://chatgpt.com/> and use the CampusConnect Project from Week 3 or start a new chat. Do not upload approved campus documents; app.py contains only code.
2. Paste the integration prompt below. Replace the HEYGEN VIDEO SOURCE placeholder with the complete embed code or normal share URL copied in Step 6.
3. Replace the VISIBLE TRANSCRIPT placeholder with the exact presenter script from Step 5. At the bottom, replace CURRENT APP.PY with your complete current app.py. Submit the prompt once.
4. ChatGPT should return one complete replacement app.py. Copy only the code inside the code block. In VS Code, replace the contents of app.py and choose File > Save.
5. The edit uses a Gradio HTML component for the disclosure, embed or link, and transcript. The official component reference is <https://www.gradio.app/docs/gradio/html>.

### **COPY / PASTE - ChatGPT integration prompt**

You are helping a beginner software-architecture student modify an EXISTING local Gradio RAG application. I will paste my current complete app.py after this prompt.

MY HEYGEN VIDEO SOURCE

<PASTE THE COMPLETE HEYGEN EMBED CODE OR SHARE URL HERE>

VISIBLE TRANSCRIPT

<PASTE THE EXACT HEYGEN PRESENTER SCRIPT FROM STEP 5 HERE>

MODIFICATION CONTRACT

1. Preserve every existing RAG feature, LM Studio URL, retrieval step, citation, abstention rule, human fallback, and `demo.launch(server_name="127.0.0.1", share=False)`.
2. Add a clearly labeled "AI Presenter" card above or beside the existing chatbot.
3. Show this visible disclosure before the video: "AI-generated presenter - not a live staff member."
4. If MY HEYGEN VIDEO SOURCE is complete iframe code, place it in `gr.HTML` without autoplay. If it is only a normal share URL, do not invent an embed URL; show a large "Open the AI presenter video" link that opens in a new tab.
5. Under the player or link, show the complete VISIBLE TRANSCRIPT inside a details/summary section labeled "Read the presenter transcript."
6. Add: "The video is optional. The text chatbot works without playing it."
7. Keep the page usable on a narrow laptop screen. Do not hide or shrink the chatbot.
8. Do not add an API key, cloud model, database, authentication, LiveAvatar, JavaScript autoplay, or unrelated feature.

Return exactly one COMPLETE replacement app.py code block, not fragments. After the code block, give only three short verification steps.

CURRENT APP.PY

<PASTE YOUR COMPLETE CURRENT APP.PY BELOW THIS LINE>



## Step 9 - Run the integrated prototype

1. In VS Code, choose Terminal > New Terminal. Confirm your Week 4 Python environment is active, then run `python app.py`.
2. Open the local Gradio URL. Confirm that the AI Presenter disclosure, embedded player or share-link fallback, transcript, and text chatbot are all visible.
3. Do not repair cosmetic details during the first run. The five tests will determine whether a change is actually needed.

## If the app does not start

1. Read the final 10-20 terminal lines. Copy only the error message; remove usernames, folder names, tokens, or private content.
2. Paste the repair prompt below into the same ChatGPT conversation. Replace both placeholders and ask for one complete `app.py`.
3. Use only one repair attempt. If it still fails, restore `app_week4_backup.py` as `app.py` and record the integration as Revise. A truthful failed prototype is valid design evidence.

### **COPY / PASTE - One permitted repair prompt**

My Week 8 local Gradio prototype did not run correctly.

ERROR OR VISIBLE PROBLEM

<PASTE ONLY THE ERROR MESSAGE OR DESCRIBE THE VISIBLE PROBLEM>

CURRENT APP.PY

<PASTE THE COMPLETE CURRENT APP.PY>

Return one complete replacement `app.py`. Preserve the original local RAG behavior, LM Studio connection, citations, abstention, and `share=False`. Keep the AI-presenter disclosure, HeyGen embed or safe link fallback, visible transcript, no autoplay, and text-chatbot fallback. Do not add an API key, hosted model, new framework, database, or unrelated feature.

## Part 4 - Run five tests and decide (9 minutes)

### TEST THE UNCERTAINTY, NOT THE POLISH

Do not score the presenter mainly on clothing, accent, or cinematic quality. Ask whether it clarifies scope and risk without preventing text-only use or breaking the Week 4 RAG behavior.

### Step 10 - Run the five exact tests

1. Test 1 - Open the page and do not press Play. Confirm that "AI-generated presenter - not a live staff member" and the transcript are visible before any audio.
2. Test 2 - Play the video once. Confirm that it lasts 30-60 seconds and states the supported topics, privacy warning, and human fallback.
3. Test 3 - Mute or pause the video. Confirm that the transcript gives the essential message and the text chatbot remains visible and usable. For why captions or transcripts matter, see [W3C's media accessibility guidance](#).
4. Test 4 - In the text chatbot, ask: How do I reset my CampusConnect password? Confirm that the original grounded answer and source citation still work.
5. Test 5 - Ask: What are the Thanksgiving dining hall hours? Confirm that the chatbot abstains or offers a safe human next step rather than inventing an answer.
6. In GitHub, edit VIDEO\_VALIDATION.md. Record one short observed result and Pass or Revise for every test.

### Worked examples: Pass versus Revise

| Observation                                                         | Decision             | Why                                                              |
|---------------------------------------------------------------------|----------------------|------------------------------------------------------------------|
| The page says AI-generated before Play; transcript is visible.      | PASS                 | A muted user receives disclosure and essential guidance.         |
| The video starts automatically with sound.                          | REVISE               | The design violates user control and mute-safe behavior.         |
| The video link opens, but the player cannot embed in the free plan. | PASS WITH LIMITATION | The explicit fallback works; record the extra click as evidence. |
| The avatar says it can reset accounts.                              | REVISE               | The script crosses the approved action boundary.                 |
| The video works, but the chatbot citation disappeared.              | REVISE               | A presentation layer must not damage the grounded RAG contract.  |

## Step 11 - Make one design decision and one small revision

1. In VIDEO\_VALIDATION.md, choose KEEP, CHANGE, or REMOVE. Base the decision on the five observations, not on whether the avatar looked impressive.
2. Make one small revision only. Examples: improve the visible disclosure, move the transcript above the fold, enlarge the fallback link, or restore a chatbot element that the code edit displaced.
3. Do not generate another video during class. If the script itself needs a future change, record it as a backlog item for the final demo.

## Part 5 - Preserve the prototype (6 minutes)

### Step 12 - Finish the record and upload the changed code

1. In VIDEO\_VALIDATION.md, complete the design decision and Engineering and AI Use Log. Name exactly what HeyGen generated, what ChatGPT changed, what you verified, and one remaining limitation.
2. In the GitHub browser, confirm feature/week-8-avatar-prototype is selected. Open docs/prototypes/week-8/VIDEO\_VALIDATION.md, click the pencil icon, paste the completed record, and commit with the message Record Week 8 validation evidence.
3. Return to the branch's Code page. Click Add file > Upload files. Upload the revised app.py. If requirements.txt changed, upload it too. Do not upload app\_week4\_backup.py, a video file, an approved campus document, .env, API key, or secret.
4. Commit with the message Add accessible AI presenter prototype. Use [GitHub's official upload instructions](#).

### Step 13 - Open the draft pull request

1. Click Compare & pull request. If the banner is absent, open Pull requests > New pull request, set base to main, and compare to feature/week-8-avatar-prototype.
2. Use the title Week 8: validate AI presenter prototype. Paste the pull-request description below.
3. Choose Create draft pull request. Do not merge it unless your instructor directs you to merge.
4. If the screen differs, use [GitHub's pull-request guide](#). Submit the draft pull-request URL.

**COPY / PASTE - Draft pull-request description**

```
## Week 8 prototype
- Added one HeyGen stock-avatar presenter path to the local CampusConnect UI
- Added visible AI disclosure and a complete transcript
- Preserved text-only use, local RAG grounding, citations, abstention, and human fallback
- Ran all five validation tests and recorded one design decision

## Safety and accessibility
- No personal digital twin, voice clone, real student data, password, or API key
- Video does not autoplay
- Text remains usable when video is muted, blocked, or ignored

## Evidence
See docs/prototypes/week-8/VIDEO_VALIDATION.md.
```

## Submission checklist

- ☐ HeyGen project uses one public stock avatar, the supplied script, and one generation.
- ☐ Presenter video or safe share-link fallback appears in the local Gradio interface.
- ☐ AI disclosure, complete transcript, no autoplay, and text-only fallback are visible.
- ☐ All five tests contain honest observations and Pass / Revise decisions.
- ☐ One KEEP / CHANGE / REMOVE decision and one small revision are recorded.
- ☐ app.py and VIDEO\_VALIDATION.md are on feature/week-8-avatar-prototype.
- ☐ No video file, personal digital twin, private campus data, .env file, or secret is in GitHub.
- ☐ One draft pull-request URL is submitted.

## Simple 100-point rubric

| Area                    | Points | Full-credit evidence                                                         |
|-------------------------|--------|------------------------------------------------------------------------------|
| Prototype contract      | 20     | Hypothesis and all seven required behaviors are explicit.                    |
| HeyGen presenter        | 20     | One stock-avatar video uses the supplied 30-60 second script.                |
| Accessible integration  | 25     | Disclosure, embed/link, transcript, no autoplay, and text fallback work.     |
| Five tests and decision | 25     | All observations are honest and support one KEEP / CHANGE / REMOVE decision. |
| GitHub trail            | 10     | Correct branch, files, commits, AI Use Log, and draft PR URL.                |

## Week 9 bridge

Week 8 deliberately produced a reversible prototype and evidence. In Week 9, Implementation, Version Control, and Code Review will ask whether the selected design deserves cleaner code, a stable component boundary, reviewable commits, and teammate scrutiny. The draft pull request is therefore not administrative paperwork; it is the handoff from design validation to implementation.